Group N · No progression recorded Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**47**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**AFR**	Pathologic T	**T1**
Prior diagnosis	**No**	Pathologic N	**N0 (I+)**
Radiation therapy	**Yes**	Pathologic M	**CM0 (I+)**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

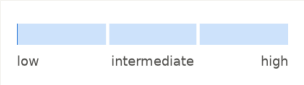
Sample: **TCGA-S3-AA14-01** · Aneuploidy score: **2** · TMB (nonsyn., /Mb): **0.53** · MSIsensor score: **0.22** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

0th

Low risk tier · fused neuro-symbolic model



Probability of progression

5% by 36 mo

12 mo 1% · 24 mo 3% · 60 mo 10%

Model-estimated progression time

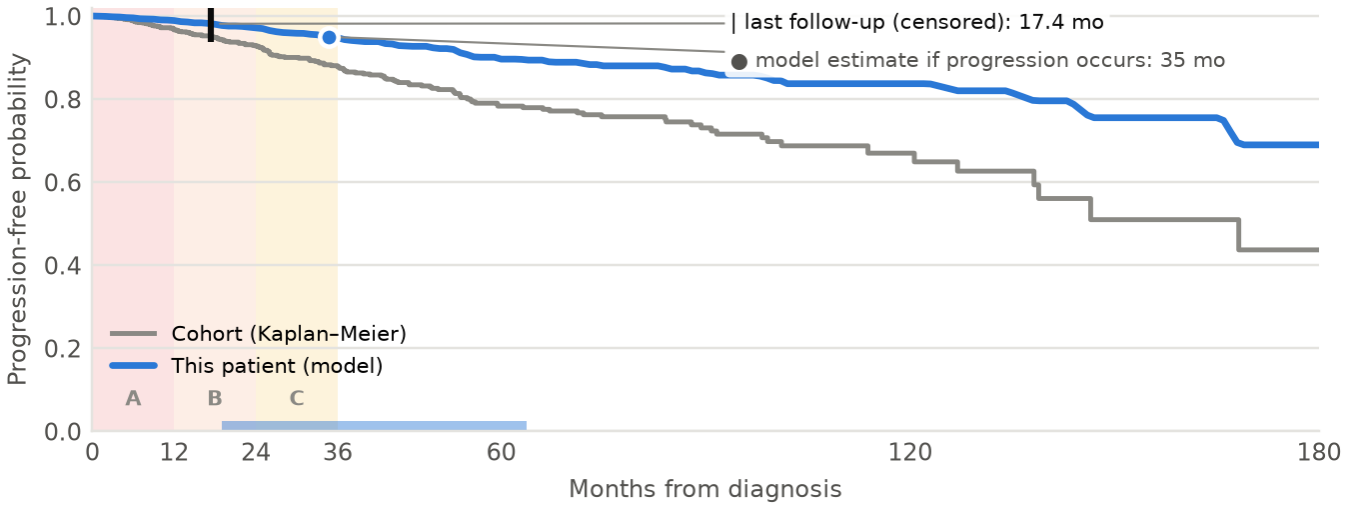
≈ 35 months

Window 24–36 months · IQR 19–64 mo
if progression occurs

Recorded progression

None recorded

Followed 17.4 mo · **Group N** · No progression recorded
Short follow-up: absence of progression is weakly informative.



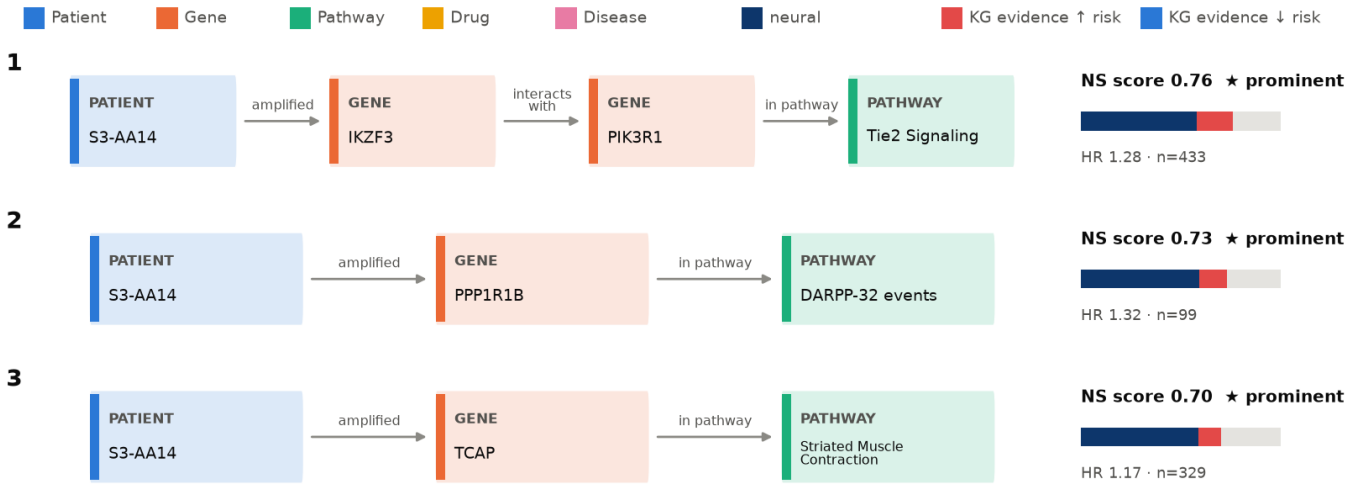
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 47-year-old patient has a LumA tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **0th risk percentile** (low tier). The most significant rule covering them (*N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **IKZF3** to **Tie2 Signaling**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.28). No progression was recorded during 17.4 mo of follow-up.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

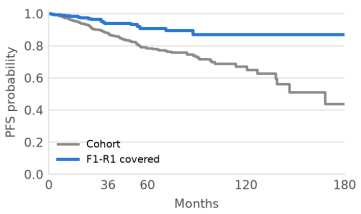


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-S3-AA14** amplified → **IKZF3** interacts with → **PIK3R1** in pathway → **Tie2 Signaling**	97	↑ **faster progression** HR 1.28, z +1.3, n = 433 carriers of “Tie2 Signaling”	**0.76**
2 ★	**TCGA-S3-AA14** amplified → **PPP1R1B** in pathway → **DARPP-32 events**	99	↑ **faster progression** HR 1.32, z +1.0, n = 99 carriers of “DARPP-32 events”	**0.73**
3 ★	**TCGA-S3-AA14** amplified → **TCAP** in pathway → **Striated Muscle Contraction**	98	↑ **faster progression** HR 1.17, z +0.8, n = 329 carriers of “Striated Muscle Contraction”	**0.70**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

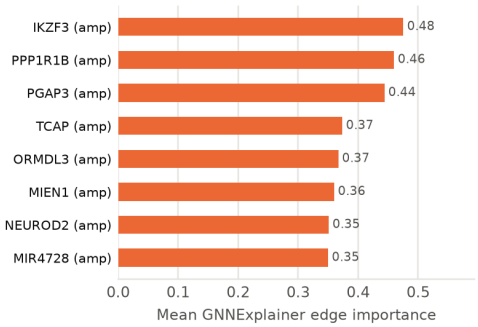
Rule	IF …	Covered	HR vs rest	Median PFS
F1-R1 ↓ lower risk	N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45	365 pts 19 events	**0.26** p 4.0e-09	not reached
F1-R7 ↓ lower risk	M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02	609 pts 61 events	**0.36** p 1.8e-08	168 mo
F1-R2 ↓ lower risk	N stage = N0 AND Age ≥ 31.5	410 pts 29 events	**0.36** p 4.5e-07	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.32
NF1 · driver	High-level amplification	GISTIC ±2	6	0.31
MAP3K1 · driver	Mutation	p.T1145Nfs*6, p.X1328_splice	13	0.31
PIK3CA · driver	Mutation	p.E542K	70	0.31
IKZF3	High-level amplification	GISTIC ±2	0	0.48
PPP1R1B	High-level amplification	GISTIC ±2	1	0.46
PGAP3	High-level amplification	GISTIC ±2	0	0.44
TCAP	High-level amplification	GISTIC ±2	1	0.37
ORMDL3	High-level amplification	GISTIC ±2	2	0.37
MIEN1	High-level amplification	GISTIC ±2	0	0.36
NEUROD2	High-level amplification	GISTIC ±2	0	0.35
MIR4728	High-level amplification	GISTIC ±2	0	0.35



19 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 29 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-3C-AALK	52	I	LumA	0.96	no progression, followed 47.6 mo
TCGA-D8-A1XM	57	I	LumA	0.89	no progression, followed 17.7 mo
TCGA-D8-A1JU	51	I	LumA	0.85	no progression, followed 14.7 mo
TCGA-D8-A1JH	56	I	LumA	0.84	no progression, followed 14.0 mo
TCGA-BH-A1ET	55	I	LumA	0.82	no progression, followed 82.8 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.